

GUSHGY, Turkmenistan

WMO: 389870

Lat: 35.283N

Lon: 62.350E

Elev: 625

StdP: 94.04

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.5	-6.5	-14.9	1.1	-5.6	-11.5	1.5	3.2	8.3	15.2	7.0	16.0	1.9	20	0.456

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	39.3	17.9	38.0	17.4	36.7	17.1	19.8	33.9	19.0	33.3	18.3	32.9	5.2	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	11.8	23.0	14.2	10.9	22.4	13.1	10.1	22.2	59.3	33.9	56.4	33.5	54.3	33.0	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	6.5	5.9	-14.0	41.9	5.1	0.9	-17.7	42.6	-20.7	43.2	-23.5	43.7	-27.2	44.4		
(5)			-14.6	21.2	5.0	1.3	-18.2	22.2	-21.1	22.9	-23.9	23.7	-27.5	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.7	4.7	6.1	11.4	16.9	22.9	27.3	29.1	27.2	21.9	16.0	9.8	6.0		
	DBStd	9.66	5.94	6.15	5.08	4.68	3.49	2.55	2.25	2.45	3.20	4.42	4.93	5.03		
	HDD10.0	571	178	132	45	6	0	0	0	0	0	7	62	142		
	HDD18.3	1820	422	342	220	79	7	0	0	0	7	98	260	384		
	CDD10.0	3006	14	23	88	213	401	519	593	534	358	192	55	17		
	CDD18.3	1213	0	1	4	36	149	269	334	276	115	25	3	1		
	CDH23.3	17166	1	4	88	443	1824	3839	5056	3931	1523	400	53	3		
	CDH26.7	9760	0	0	18	158	879	2298	3206	2361	692	136	12	0		
(14) Wind	WSAvg	2.6	2.2	2.4	2.4	2.3	2.6	3.2	3.6	3.2	2.6	2.2	2.2	2.0		
(15) Precipitation	PrecAvg	292	49	55	75	43	15	0	0	0	0	2	13	40		
	PrecMax	445	123	121	168	129	72	6	0	0	7	16	48	136		
	PrecMin	156	5	10	14	1	0	0	0	0	0	0	0	0		
	PrecStd	84	30	32	42	30	18	1	0	0	1	4	13	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.7	23.5	29.3	33.8	38.0	40.2	41.2	40.1	36.6	33.8	29.0	23.1		
		MCWB	10.5	11.4	14.1	16.3	17.2	18.1	18.7	18.1	16.7	14.8	13.6	10.4		
	2%	DB	18.9	20.6	26.0	30.8	35.4	38.4	39.7	38.1	34.3	30.4	24.6	19.9		
		MCWB	9.2	10.8	13.2	16.3	16.9	17.6	18.1	17.1	15.8	14.3	12.0	9.3		
	5%	DB	16.1	17.8	23.0	28.3	33.5	37.0	38.4	36.8	32.7	27.8	21.5	17.0		
		MCWB	8.2	9.5	12.9	15.7	16.5	17.3	17.8	16.9	15.3	13.8	11.0	8.6		
	10%	DB	13.3	15.3	20.1	25.7	31.5	35.6	37.1	35.5	30.8	25.2	18.6	14.1		
		MCWB	7.4	9.0	12.3	15.2	16.1	16.8	17.3	16.5	14.8	13.0	10.2	7.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.0	13.3	16.7	20.2	19.7	20.2	21.1	20.0	18.4	16.4	14.5	11.8		
		MCDB	18.6	19.7	24.1	26.4	30.8	35.4	35.1	33.8	32.1	28.9	24.6	18.7		
	2%	WB	10.2	11.9	14.9	18.2	18.5	19.0	20.0	18.9	17.2	15.3	13.1	10.4		
		MCDB	16.7	18.1	21.8	25.3	30.2	34.5	35.4	33.7	30.7	27.3	22.0	17.6		
	5%	WB	8.8	10.7	13.8	16.9	17.7	18.3	19.1	18.1	16.3	14.5	12.1	9.2		
		MCDB	14.8	16.3	20.5	24.8	29.6	33.7	34.7	33.8	30.0	25.9	19.4	15.6		
	10%	WB	7.6	9.4	12.7	15.8	17.1	17.7	18.3	17.3	15.5	13.6	10.9	8.0		
		MCDB	12.8	14.2	19.3	24.0	28.6	32.8	34.5	33.5	29.2	24.1	17.3	13.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	10.2	11.3	12.4	14.8	15.4	15.2	15.9	16.6	15.4	12.2	10.3		
		MCDBR	15.4	15.1	17.1	17.6	18.2	17.6	16.6	17.0	18.6	19.2	17.5	15.9		
	5% WB	MCWBR	7.5	7.1	6.9	6.4	5.8	5.7	5.1	5.5	7.4	8.5	8.0	7.4		
		MCWBR	13.3	13.3	14.0	14.0	15.0	15.0	14.3	14.6	16.4	16.7	14.5	13.5		
(40) Clear-Sky Solar Irradiance	taub		0.358	0.406	0.473	0.502	0.506	0.486	0.469	0.434	0.401	0.424	0.404	0.356		
	taud		2.174	2.053	1.899	1.873	1.878	1.917	1.963	2.043	2.101	2.010	2.057	2.185		
	Ebn at Noon		818	816	791	791	793	808	818	841	850	783	752	792		
	Edh at Noon		114	145	184	199	200	192	183	165	148	148	127	106		
(44) All-Sky Solar Radiation	RadAvg		2.41	3.16	4.38	5.73	7.35	8.28	8.05	7.44	6.32	4.64	3.04	2.30		
	RadStd		0.24	0.48	0.42	0.58	0.32	0.18	0.12	0.14	0.12	0.23	0.30	0.28		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	+0.53	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page